ERC Advanced Grant 2026
Part B2

1 Research Methodology

We propose **Co-Evolutionary Personalised AI (CoPAI)**, a multi-agent framework in which the AI system and the human user co-evolve. CoPAI is built on an information-theoretic framework that replaces preference-based reward with **learnable information gain**. The framework combines a diagnostic system with a learning algorithm called **Learnable Information Policy Optimisation (LIPO)**. The diagnostic system tracks whether human–AI interaction is productive (e.g., users gain independent capability) or harmful (e.g., users become dependent or stagnant). LIPO then adapts the AI's support strategy to the diagnosed situation. CoPAI is organised around four pillars, **Measure**, **Model**, **Steer**, and **Validate**, which are instantiated as 5 Work Packages (WPs) as shown in Figure 1. WP1 develops measures of human capability growth, AI agent personalisation gains, how closely humans and AI work together, how human's roles shift, and whether learning carries beyond tasks at hand. WP2 builds a user model that tracks cognitive state, extracts personal learning patterns, and develops a hierarchical memory system with generative working memory. WP3 creates a world model that predicts how users respond to AI actions and how their cognitive state evolves. WP4 develops the personalised policy together with LIPO that optimises for long-term human capability growth. WP5 evaluates CoPAI through a long-term education trial comparing it with satisfaction-based personalisation and a non-personalised baseline, and releases CogWell-Bench for assessing autonomy, dependency, and wellbeing in personalised AI.

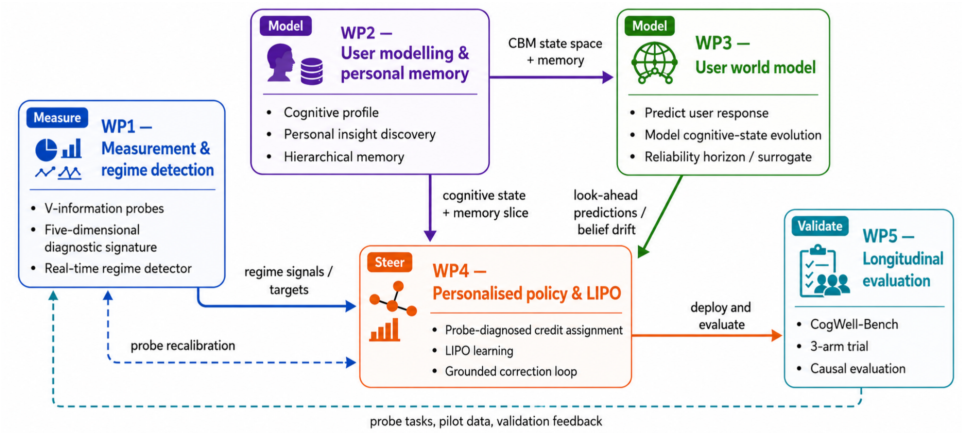

Figure 1: Work package dependencies.

WP1: $\mathcal{V}$-Information Measurement and Regime Detection (M1–24)

Pillar I (Measure). Lead: PI & PDRA-1; contributors: PDRA-2, PhD-1.

Goal. Design, calibrate, and validate **$\mathcal{V}$-information probes** that measure human–AI interaction; and develop a **regime detector** that classifies the co-evolutionary health of each human–AI trajectory.

Novelty. WP1 advances cognitive measurement in human–AI interaction: **(i)** It introduces *$\mathcal{V}$-information [1] probes for co-evolutionary health*. To our best knowledge, it is the first operationalisation of information-theoretic diagnostics for longitudinal, individual-level human–AI trajectories; and **(ii)** It provides a *co-evolutionary regime detector* that identifies healthy human–AI co-evolution and failure modes as they develop.

Overview. Direct assessment gives a grounded measure of what learners can do without AI support. However, it can only be used occasionally, because frequent testing may disrupt learning or cause frustration. Assessment scores only show which answers are wrong, but not *why* they are wrong. The same score may reflect different cognitive states, such as a misconception, a change in reasoning strategy, productive confusion, or learned helplessness. These cases require different pedagogical actions. More importantly, to learn an effective AI tutoring policy model, we need dense rewards to drive effective pedagogical policy learning [2, 3, 4].

Task 1.1 (M1–12): $\mathcal{V}$-Information Probe Design and Calibration. *Can co-evolutionary health be measured at all?* This task builds and calibrates the $\mathcal{V}$-information probes that estimate the five-dimensional signature $(\Delta\mathcal{E}^H, \Delta\mathcal{E}^A, C, \rho, \Delta_{\text{far}})$ from human–AI interaction data. For the human side, we operationalise $I_{\mathcal{V}}(H_t \to y)\big|_{\mathcal{T}}$ using lightweight probe classifiers trained on unassisted task responses, following the finite-sample $\mathcal{V}$-information protocol of [1, 5, 6]. The probe predicts held-out task success $\hat{y}$ from user cognitive state and task:

$$p(\hat{y} \mid \mathbf{c}_u, \text{task}_i) \;=\; f_{\mathcal{V}}(\mathbf{c}_u, \text{task}_i), \tag{1}$$

where $\mathbf{c}_u$ is the cognitive state for user u estimated by the User Cognitive Agent (UCA, WP2), which maps interaction trajectories to grounded psychometric dimensions (e.g., mastery, reasoning style, epistemic state, learning dynamics). The probe family is selected by held-out predictive log-likelihood on pilot data and fixed within each probe window. Cognitive growth over an interval $[t_0, t_1]$ is the change in probe's predictive capabilities:

$$\Delta\mathcal{E}^H_{[t_0,t_1]} = I_\mathcal{V}(H_{t_1} \to y)\big|_\mathcal{T} - I_\mathcal{V}(H_{t_0} \to y)\big|_\mathcal{T}. \tag{2}$$

Because the probe takes the UCA-estimated cognitive state as input, $\Delta\mathcal{E}^H$ can be decomposed by cognitive dimensions, yielding concept-level growth profiles that feed WP4 credit assignment (linking turns to concept-level gains) and T1.2 regime discovery (identifying which cognitive dimensions grow, stagnate, or regress). The AI-side term $I_\mathcal{V}(A_t \to f)\big|_\mathcal{P}$ is estimated by contrasting a null probe (predicting learner feedback f without seeing A_t) against an informed probe. The coupling term C is identified via randomised adaptive-vs-static tutor comparisons, and the unobservable frozen-human component is treated as a User World Model (UWM) simulation diagnostic (WP3). The probe task $\mathcal{T}$ (built in WP5) spans in-distribution, near-transfer, and far-transfer items, with Δ_d (for $d \in \{\text{in}, \text{near}, \text{far}\}$) [7] obtained by stratifying $\Delta\mathcal{E}^H$ across them. The frequency of probing task assessment is calibrated in the WP5 pilot, initially targeting every 5–10 sessions to balance testing effects, fatigue, and credit-assignment resolution.

Task 1.2 (M10–18): Regime Discovery. *Do distinct co-evolutionary regimes exist, and do they matter?* We discover regimes from data rather than predefining regime categories, and drop them if they prove spurious or useless to the policy in WP4. We perform clustering on a short windowed segment of the signature trajectory that captures recent trend of five-dimensional signature measures. Evolutionary clustering [8, 9] is employed to ensure that assignments remain reasonably stable across consecutive windows. A nonparametric mixture model [10] allows the number of regimes to emerge from the data and returns probabilistic assignments. The resulting assignment sequences will also be used to estimate likely transitions between regimes and provide one-step forecasts for the T1.3 early-warning analysis and WP4 policy gate. Reliability is checked with held-out expert annotation used *post hoc* to confirm clusters are interpretable, and the discovered structure is compared against the hypothesised regimes in the Part B1 table. A close match supports the hypotheses, a divergent one may reveal more informative patterns. After stable regimes are identified, we train a lightweight detector to assign new incoming sessions to regimes from short 5–10 session windows. The detector is supervised using the discovered cluster labels and reports calibrated uncertainty, so the policy can account for ambiguous cases.

Task 1.3 (M15–24): Longitudinal Probe Validation. *Does the signature reliably track regimes early enough to act?* Using WP5 pilot data over 3–6 months, we test: (i) *convergent validity*—the signature trajectory correlates with independent assessments; (ii) *discriminant validity*—regime classification predicts qualitatively different future outcomes; and (iii) *early-warning sensitivity*—Regime assignment identifies trajectories moving toward dependency or a sudden drop in performance earlier than course-embedded assessment scores do.

Deliverables. D1.1: Calibrated $\mathcal{V}$-information probe library and estimators (M12). D1.2: Real-time regime detector (M18). D1.3: Longitudinal validation of regime detection against independent assessments (M24).

WP2: User Cognitive Modelling and Personal Memory (M7–30)

Pillar II (Model). Lead: PI & PDRA-2; contributors: PDRA-1, PhD-1.

Goal. Develop a **latent user state model** that tracks the user's evolving cognitive or behavioural states in human–AI interaction; develop an **insight-extraction module** that discovers transferable patterns from accumulated interaction trajectories; and build **hierarchical personal memory** with generative working memory.

Novelty. Current personalised AI systems treat the user as a static preference profile or a short-context dialogue partner. CoPAI models the user as a *dynamical system* whose latent state evolves across interactions and whose trajectory contains extractable regularities. Our novelties: **(i)** A Concept Bottleneck Model (CBM) grounds the user representation to established psychometric constructs; and **(ii)** a personalised memory system building on our xMemory framework [11] that selectively retains and retrieves state-relevant information across sessions without unbounded memory growth.

Task 2.1 (M7–12): Cognitive Concept Bottleneck for latent user state modelling. *What does the system need to know about each user?* The User Cognitive Agent (UCA) models each user's cognitive profile $\mathbf{c}_u$ as a structured latent combining theory-grounded and data-driven dimensions. The Cognitive Profile Space $\mathcal{C}$ decomposes into two subspaces. A *grounded subspace*, estimated via a Concept Bottleneck Model [12] trained on psychometrically labelled interaction data (T2.4), partitions for education into mastery, reasoning-style, epistemic-state, and learning-dynamics blocks, ensuring alignment with established constructs. A *residual subspace*, conditioned on the grounded part and learned via our user latent-factor modelling [13, 14], captures

AI-interaction-specific patterns existing instruments miss (e.g. scaffolding-style sensitivity). The residual dimensions are analysed post-hoc via probing classifiers to assess their interpretability. The full user profile separates stable *traits* from time-varying *states*. Trait components feed directly into T2.3's semantic memory layer; state components are reconstructed each turn in T2.3's working memory layer.

Task 2.2 (M13–18): Personal Insight Discovery (PID) via experiential knowledge extraction. *How does the system turn raw interaction into reusable knowledge?* PID extracts structured, reusable insights from trajectory batches, e.g., misconception clusters, transfer-difficulty profiles, and zone-of-proximal-development (ZPD) boundary estimates, following our metacognitive consolidation framework [15]. The extractor is an LLM with structured output, trained via on-policy context distillation [16] to generalise from few observations. Within the memory hierarchy (T2.3), insights serve two purposes. First, they are stored as structured episodic entries that can be retrieved when needed. Second, they support long-term consolidation. Once an insight is confirmed across enough batches, it is converted into a semantic summary by adapting our context-compression method [17], and the original episodes are removed to keep memory use manageable.

Task 2.3 (M16–24): Personal hierarchical memory with generative working memory. *How to construct personal memory, and how is it used at inference?* We develop a three-layer memory structure $\mathcal{M} = (\mathcal{M}^{\text{sem}}, \mathcal{M}^{\text{epi}}, \mathcal{M}^{\text{wm}})$. The *semantic layer* $\mathcal{M}^{\text{sem}}$ encodes stable trait knowledge in a low-rank adapter, updated only under consistent contradictory evidence [18, 19]. The *episodic layer* $\mathcal{M}^{\text{epi}}$ stores interaction memories, decoupled into semantic components via our xMemory framework [11], alongside PID insights. The key novelty is the *working-memory layer* $\mathcal{M}^{\text{wm}}$. Rather than retrieving stored chunks, it *generates* a situated cognitive context, conditioning on the full profile $\mathbf{c}_u$ and episodic anchors ranked by our Spectrum Projection Score [20], which selects memories by downstream reasoner utility rather than surface similarity.

Task 2.4 (M25–30): Psychometric Validation. *Is cognition recoverable from interaction at all?* We will start with our existing personalised AI tutoring system LearnLens [21]. Human-subject studies (at least 200 participants) provide ground truth for cognitive profiling. Participants complete standardised psychometric instruments (e.g., Cognitive Style Index, Need for Cognition Scale, Rational–Experiential Inventory), interact with LearnLens across 10+ sessions, and complete held-out cognitive tasks. Ethics approval is initiated at Month 1, ensuring data collection begins no later than Month 6. We test the hypothesis that computationally inferred profiles predict psychometric scores above chance, validating that cognition is recoverable from interaction data; negative results would characterise identifiability limits. The validation data also calibrates the co-evolution learning algorithm in **WP4** and establishes the pre-intervention cognitive baselines required for evaluation in **WP5**.

Deliverables. D2.1: UCA for user cognitive profiling (M12). D2.2: PID insight extractor and downstream validation (M18). D2.3: Hierarchical personal memory with generative working memory (M24). D2.4: Validated cognitive profiling system with psychometric correlation results (M30).

WP3: User World Model (UWM) of Cognitive Evolution (M13–36)

Pillar II (Model). Lead: PI & PDRA-2; Contributors: PDRA-1, PDRA-2, PhD-2.

Goal. Learn a UWM that simulates how individual users respond to AI actions and how their cognitive state evolves over interaction trajectories, providing (i) look-ahead value estimates that let the reasoner (WP4) evaluate candidate actions before committing, and (ii) a fast user simulator for self-play training (WP4).

Novelty. Treating the human user as the environment whose cognitive state changes in response to AI actions [22], this WP makes two contributions: **(i)** It build a *User World Model* (UWM) that estimates the user's cognitive state and predicts their likely response, extending our Bayesian inverse planning framework [23]; and **(ii)** It supports data-efficient personalisation by combining shared population-level patterns with user-specific adaptation, using regime-aware aggregation to preserve differences across user types. This extends our PROPER framework [24] for personalised LLM learning to world-model personalisation.

Task 3.1 (M13–21): User World Model (UWM) through Bayesian Inverse Planning. *Can we predict how a user will change, not just where they are now?* WP2 gives a static estimate $\mathbf{c}_u^{\text{CBM}}$ of current cognitive state. WP3 tracks how the user evolves and predicts future behaviour while staying calibrated to WP2's Concept Bottleneck Model (CBM). The tutor is proactive: its action a_t poses the next task or prompt, and the learner's reply o_t is the observation from which state is inferred. The UWM has two components. The first is a *cognitive-state inference* module infers a user's latent state from their behaviour by Bayesian inverse planning, instantiating Bayesian inverse planning [25] over cognitive rather than goal states, extending our own application of this framework to latent mental state inference in dialogue [23]. At each turn, it observes the user's response o_t to the AI tutor action a_t and updates its posterior over the cognitive state via Bayesian inversion:

$$p(\mathbf{c}_{t+1} \mid o_{1:t}, a_{1:t}) = \int \underbrace{p(\mathbf{c}_{t+1} \mid \mathbf{c}_t, o_t, a_t)}_{\text{transition}} \underbrace{p_{\text{LLM}}(o_t \mid \mathbf{c}_t, a_t)}_{\text{likelihood}} \underbrace{b_t(\mathbf{c}_t)}_{\text{prior}} \, \mathrm{d}\mathbf{c}_t, \tag{3}$$

where $b_t(\mathbf{c}_t) \triangleq p(\mathbf{c}_t \mid o_{1:t-1}, a_{1:t-1})$ is the recursive prior summarising all previous turns. The likelihood $p_{\text{LLM}}(o_t \mid \mathbf{c}_t, a_t)$ is implemented by the LLM backbone, which scores how probable the observed learner reply is under each candidate cognitive state. The posterior is tracked within WP2's interpretable CBM concept space, so the system can show what it learned about the user after each interaction. The second component is a *response-generation* module then predicts observable behaviour (e.g., answer correctness, error type, latency, help-seeking) condition on the current user cognitive state posterior. Training jointly maximises next-response accuracy and calibration to WP2's independent CBM estimate, allowing flexibility without forcing exact agreement. Our approach extends Bayesian Knowledge Tracing [26] from binary skill mastery to continuous, interpretable, action-conditioned cognitive profiles. It will be trained on LearnLens data and evaluated against direct prediction and Deep Knowledge Tracing [27] baselines.

Task 3.2 (M19–24): Amortised Surrogate and Reliability Self-Diagnostic. *Can the model run fast enough for self-play, and know when to trust itself?* The full UWM is costly since each update (Eq. 3) requires scoring many candidate states through the LLM likelihood, too slow for the rollouts WP4 self-play needs. We therefore distil an *amortised surrogate* $q_\phi(\mathbf{c}_{t+1} \mid o_{1:t}, a_{1:t})$ that approximates the full model's posterior in a single forward pass from the UWM's own outputs (no ground-truth state labels required), structurally similar to our self-distillation approach [28]. The surrogate is the fast simulator for WP4, while the full model remains the reference for evaluation and reward calibration. The surrogate also includes a *reliability self-diagnostic*. When its uncertainty, or its disagreement with the full model on sampled turns, exceeds a calibrated threshold, the system falls back to the full UWM or to observed signals only. The surrogate also estimates how far ahead it can predict reliably. The same self-check produces a reliability score that guides how WP4 assigns credit. When the system is unsure, credit is reduced in a safe and cautious way.

Task 3.3 (M21–36): World-Model Personalisation and Continual Learning. *How does a population-level user world model adapt to each individual user?* The population-level UWM learns patterns that are shared across users, such as which types of AI actions tend to affect which cognitive concepts and how cognitive states usually change over time. This is then adapted to individual cognitive states, transition patterns, and response styles. We extend our progressive-personalisation framework PROPER [24] from preference adaptation to world modelling, adaptation proceeds population $\to$ group (users clustered by CBM-profile similarity) $\to$ individual as data accumulates. As the deployed user population becomes more diverse, the population-, group-, and user-level parameters need to adapt to new users while maintaining performance for users already served. This is the main stability–plasticity challenge in personalisation, learning new patterns without forgetting existing ones [29]. We plan to explore different approaches to address this [30]. One potential way is to first estimate which parameters are most important for previously served CBM clusters. Updates to these important parameters are reduced to preserve existing personalisation knowledge, while less critical parameters are adapted to new user groups, following parameter-importance methods [31, 32, 33]. We could also construct a replay buffer which ensures balanced coverage across user regimes discovered in WP1, since users from different co-evolutionary regimes follow different behavioural patterns, adapting replay-based continual learning approaches [34, 35].

Deliverables. D3.1: User World Model (UWM) that tracks users' cognitive states over time and predicts their future behaviour (M21). D3.2: An efficient surrogate model that supports simulation and planning while detecting when its predictions are unreliable (M24). D3.3: Personalised UWM that adapts to individual users and improves over time without forgetting (M36).

WP4: Personalised Policy and Co-Evolutionary Learning (M19–52)

Pillar III (Steer). Lead: PI, PDRA-3; Contributors: PDRA-2, PhD-3.

Goal. Develop the personalised policy together with its co-evolutionary learning algorithm, LIPO (Learnable Information Policy Optimisation), that optimises for long-term human capability growth.

Novelty. LIPO shifts personalised AI from optimising what users prefer in the moment to optimising how users build capability over time. Its novelties are: **(i)** it replaces preference or engagement reward with a per-turn signal obtained by *credit assignment of a measured capability outcome*, the trajectory-level $\mathcal{V}$-information gain $\Delta\mathcal{E}^H$; and **(ii)** it uses delayed grounded evidence to correct both the policy and the diagnostic that produced its reward.

Overview. A clear answer may satisfy learners, but it may also remove the productive struggle that helps them build real skill [36, 37, 38]. LIPO therefore distinguishes an *interaction reward* r_{int} (engagement, satisfaction; dense, per-turn) from a *grounded reward* r_{gnd} (independent problem-solving after support is removed; delayed, observed only at probes). WP4 proceeds in four stages. T4.1 constructs the training signal, redistributing each realised return across the turns that produced it. T4.2 trains the policy against the WP3 User World Model (UWM) through self-play. T4.3 then corrects what simulation gets wrong, using delayed probe outcomes to update both the policy and the WP1 diagnostic. T4.4 evaluates LIPO and whether it transfers beyond education.

Constrained objective. We formulate a constrained belief-state Markov Decision Process (MDP) with augmented state $s_t = (\mathbf{c}_t, \boldsymbol{\gamma}_t, m_t)$, comprising WP2 cognitive state $\mathbf{c}_t$, WP1 regime classification distribution $\boldsymbol{\gamma}_t$, and personal-memory slice m_t. We write $\boldsymbol{\gamma}_t \in \Delta^{K-1}$ for the regime detector's calibrated posterior over the K regimes. Treating the inferred regime as part of the policy state (rather than re-shaping the reward as regimes vary) preserves a stationary MDP:

$$\max_{\pi} \ \mathbb{E}_{\pi}\Big[\textstyle\sum_t \delta^t r_{\text{gnd},t}\Big] \quad \text{s.t.} \quad \mathbb{E}_{\pi}\Big[\textstyle\sum_t \delta^t r_{\text{int},t}\Big] \geq \tau, \tag{4}$$

where $\delta \in (0, 1)$ is the discount factor, and τ is the minimum acceptable interaction quality (the system must stay usable) [39, 40]. Since r_{gnd} is observed only at probes, LIPO trains on the dense credited reward of T4.1 and calibrates it offline against grounded outcomes in T4.3. The system enforces a minimum interaction quality level (through r_{int}) by adding a penalty during the T4.2/T4.3 policy updates when quality falls below the target.

Task 4.1 (M19–28): Probe-diagnosed Credit Assignment of the $\mathcal{V}$-information Return. *How do we turn a sparse capability measure into a dense training signal?* We treat the realised $\Delta\mathcal{E}^{H}_{[t_0,t_1]}$ over a probe interval as a coarse grounded signal and redistribute it across turns, extending the evidence-diagnosed step localisation of our EDIT framework [41] from token credit within one generation to turn-level credit across sessions. Each turn receives a share of the total change based on its relative contribution, so credit is redistributed [42, 43]. The contribution of each turn depends on three factors: (1) how much the model's belief about the learner changed at that turn (measured by the Hellinger distance between consecutive UWM cognitive-state estimates), (2) how uncertain the probe was at that point (giving more weight to challenging turns that later lead to learning gains), and (3) the reliability of the UWM estimate. Because the belief changes are represented in the CBM concept space, attribution is linked to specific cognitive constructs rather than an unexplained score. If UWM reliability decreases, the weighting becomes more uniform, gradually falling back to equal credit across turns. In the extreme case, attribution relies only on the grounded probe outcome $\Delta\mathcal{E}^{H}$, similar to NOVER-style verifier-free outcome filtering [44]. The credited reward measures capability gain but does not capture whether that gain comes with increasing dependence on AI. We therefore adjust the reward using a reliability-aware zone of proximal development (ZPD) check and a dependency-risk penalty. The ZPD constraint is stronger when the learner model is reliable and relaxes when uncertainty is high. The dependency penalty uses WP1 signals on coupling and role balance, with stronger penalties for dependency risks and weaker ones for productive collaboration. This adjusted reward is used by both T4.2 and T4.3.

Task 4.2 (M27–38): Self-Play Policy Training under the User World Model. *How does the AI system learn when real interaction data is scarce?* Real deployment yields far too few trajectories to train a policy over long horizons. A single learner produces perhaps a hundred sessions across a semester, and the capability signal arrives only at probes. T4.2 therefore trains the policy against the amortised surrogate q_ϕ (D3.2), which simulates learner responses and cognitive-state evolution at rollout speed. The policy is parameterised as a shared representation with lightweight regime-specific heads,

$$\pi_\theta(a_t \mid s_t) \ = \ \sum_{k=1}^{K} g_k(\boldsymbol{\gamma}_t, \, \hat{\boldsymbol{\gamma}}_{t+1}) \ \pi_{\theta_k}(a_t \mid \mathbf{c}_t, m_t), \tag{5}$$

where $\hat{\boldsymbol{\gamma}}_{t+1}$ is the one-step-ahead regime forecast from T1.2, and $g : \Delta^{K-1} \times \Delta^{K-1} \to \Delta^{K-1}$ is a learned gating function whose k-th component $g_k(\cdot, \cdot) \in [0, 1]$ returns the mixing weight for head k, with $\sum_k g_k = 1$. The policy thus conditions on *trajectory*, not current state alone. For example, stable productive struggle can lead to different support than productive struggle that appears to be shifting toward dependency. If no reliable regime structure is found, or if T4.4 shows no benefit for far transfer, the model falls back to a single-head flat policy trained on the same credited reward. The policy controls the AI system's pedagogical strategy. The action space includes hint provision, scaffolding level, diagnostic questioning, self-explanation prompts, error handling strategy, control transfer, and task choice. This allows the system to balance immediate support with long-term learner independence. Training uses asymmetric self-play. Only interactions predicted to produce meaningful learner-state change are retained for training; uninformative interactions or those where the AI system solves the problem entirely are discarded [45].

Task 4.3 (M32–42): Policy Training with Grounded Correction. *How does delayed ground truth correct both the policy and the measure that trained it?* T4.2 trains the policy model through synthetic data generated from self-play. But it needs to be validated on real probes. T4.3 uses grounded probe outcomes to update both the policy and the diagnostic system. After each probe cycle, trajectories are labelled with grounded outcomes and credited rewards are re-signed. Inspired by NOVER [44], the update combines a policy correction (weighted supervised fine-tuning on trajectories filtered by grounded probe outcomes) and a shaping correction targeting the diagnostic itself (turns where high predicted belief drift was contradicted by poor outcomes are flagged, giving a gradient that recalibrates the WP1 probe and β_k). This closes the loop: WP1 diagnostics $\rightarrow$ T4.1 credited reward $\rightarrow$ grounded validation $\rightarrow$ WP1 recalibration. When $\Delta\mathcal{E}^H$ stays below threshold for a number of sessions, the UWM identifies high-gain regions and biases pedagogical action selection, an analogue of ensuring learnable information gain in self-play [45]. We adopt three-timescale stochastic approximation [46]: probe and UWM update slowest, the regime classifier at intermediate rate, the policy fastest.

Task 4.4 (M43–52): Ablation and Cross-domain Instantiation. *Do regimes actually help, and does the framework transfer?* We evaluate four variants on LearnLens: (a) non-personalised baseline; (b) credit-assigned shaping, flat policy (T4.1); (c) regime-conditioned policy through self-play (T4.2); (d) full LIPO with per-regime heads and grounded correction (T4.1–4.3). We also plan to test whether CoPAI generalises beyond education by specifying an instantiation in health behaviour change on a personalised health-assistant setting [47]. We identify which components require domain re-specification. For example, probe tasks shift from curriculum activities to self-management scenarios; the CBM concept space is redefined around health behaviours such as self-efficacy and planning. The framework is evaluated offline using our existing health-platform interaction logs, testing whether the five-dimensional signature can be measured and whether the regime structure discovered from T1.2 also appear in this domain. Health trial is deferred to future work beyond this project.

Deliverables. D4.1: Algorithm for turn-level credit assignment based on the trajectory- or session-level $\mathcal{V}$-information gains (M28). D4.2: Self-play training pipeline for the personalised policy under the WP3 User World Model (M34). D4.3: Grounded correction loop updating both policy and WP1 diagnostic from delayed probe outcomes (M42). D4.4: Ablation study over credit assignment, regime conditioning, and grounded correction (M52). D4.5: Health-domain instantiation of CoPAI (M52).

WP5: Longitudinal Evaluation (M7–60)

Pillar IV (Validate). Lead: PI, PDRA-3; Contributors: PDRA-1, PDRA-2, PhD-2, PhD-3.

Goal. Develop and validate a longitudinal evaluation framework for personalised AI systems, testing whether AI support produces *internalised human capability* rather than short-term scaffolded performance.

Contributions. Our Contributions are: **(i)** We evaluate personalised AI using behavioural and cognitive outcomes rather than self-report satisfaction alone; and **(ii)** Building on our personalised AI tutoring work [48, 21], we release CogWell-Bench, which operationalises cognitive autonomy, dependency, behaviour change, and wellbeing as measurable benchmark dimensions, linking interaction-level features (e.g., scaffolding density, reasoning-strategy selection, Goldilocks-zone targeting) to trajectory-level outcomes.

Task 5.1 (M7–24): CogWell-Bench v0 Design and Pilot Validation. *Can co-evolutionary health be measured reliably enough to evaluate?* CogWell-Bench defines the evaluation dimensions such as preference-following accuracy, autonomy preservation, dependency trajectory, longitudinal capability change, combining automated metrics (from interaction logs and WP2 state estimates) with expert-rated measures (reasoning quality, self-explanation depth; rubrics targeting inter-rater reliability $\kappa \geq 0.7$). Ethics preparation begins at Month 1, enabling LearnLens collection from Month 6. A 4–6 week pilot ($n \approx 80$) will validate the measures, calibrate the WP1 mastery and transfer probes, estimate attrition, and support pre-registration of the trial protocol.

Task 5.2 (M37–56): Longitudinal Trial. *Does optimising for capability beat optimising for satisfaction?* We run a longitudinal education trial with phased cohort rollout. Participants are randomised within cohort to: (a) a non-personalised baseline with generic support; (b) a personalised system designed to maximise user satisfaction and perceived helpfulness; or (c) CoPAI. All conditions share the same base model family, factual resources, retrieval systems, and safety safeguards, isolating the effect of the personalisation objective. Safety is monitored through three safeguards: WP3 fallback when the system is uncertain, WP4 checks for rising user dependency, and standard procedures for handling distress. Across two semesters, we recruit LearnLens users, targeting $n \geq 200$ per condition. Assessments occur at baseline, midpoint, semester end, and 30-day follow-up. The primary endpoint is delayed solo transfer performance, normalised by baseline ability; secondary outcomes include CBM mastery growth, misconception repair, metacognitive calibration, self-regulated learning, self-explanation quality, and satisfaction. Note that the primary endpoint uses a separate assessment bank not involved in WP1 probe training or T4.3 recalibration. Probe and endpoint items come from independently developed and validated pools.

Task 5.3 (M53–60): Trial Analysis and CogWell-Bench v1 Release. *What drives the effect, for whom, and at what cost?* We analyse which CoPAI components drive outcomes, whether effects are equitable across learner groups, and how CoPAI compares on cost-effectiveness with the satisfaction-optimised and non-personalised arms. Cross-domain transfer is evaluated on the WP4 health-assistant log data. CogWell-Bench v1.0 (M60) releases the evaluation criteria, documentation, baselines, aggregate trial statistics, and controlled-access de-identified data. Where direct release is restricted, we provide synthetic data and reproducible evaluation scripts.

Deliverables. D5.1: CogWell-Bench v0, pilot reliability report, and trial protocol (M24). D5.2: Three-arm trial results (M56). D5.3: Trial findings report and CogWell-Bench v1 release (M60).

2 Workplan

The project will be carried out in 5 work packages (WPs) within 60 months. Three full-time PDRAs will be hired, each for 36 months, with staggered start dates of Year 1, 2 and 3. Three PhD students will provide depth across the key novelty pillars (WP1—Measure, WP2/3—Model, WP4—Steer). The project will fund two students, with KCL supporting the third. PhD-1 (Years 1–4) will develop $\mathcal{V}$-information diagnostics and user cognitive modelling; PhD-2 (Years 1–4) will focus on user world modelling; and PhD-3 (Years 2–5) on co-evolutionary learning. The WPs/tasks, the time allocation of PDRAs in terms of person-months, and timing of deliverables are shown in the Gantt chart below.

	WP/Task	RA1	RA2	RA3	Total	Year 1		Year 2		Year 3		Year 4		Year 5	
						H1	H2	H1	H2	H1	H2	H1	H2	H1	H2
WP1	T1.1 V-Information Probe Design and Calibration	12	0	0	12										
	T1.2 Regime Discovery	5	1	0	6										
	T1.3 Longitudinal Probe Validation	4	1	0	5										
WP2	T2.1 Cognitive Concept Bottleneck	6	0	0	6										
	T2.2 Personal Insight Discovery	2	2	0	4										
	T2.3 Personal Hierarchical Memory	1	5	0	6										
	T2.4 Psychometric Validation	1	3	0	4										
WP3	T3.1 User World Model through BIP	2	4	0	6										
	T3.2 Amortised Surrogate and Self-Diagnostic	1	5	0	6										
	T3.3 World Model Personalisation	1	5	1	7										
WP4	T4.1 Probe-diagnosed Credit Assignment	0	1	5	6										
	T4.2 Self-Play Policy Training	0	2	4	6										
	T4.3 Policy Training with Grounded Correction	0	2	5	7										
	T4.4 Ablation and Cross-domain Instantiation	0	1	5	6										
WP5	T5.1 CogWell-Bench v0 Design / Pilot Validation	1	2	4	7										
	T5.2 Longitudinal Trial	0	2	6	8										
	T5.3 Results Analysis / CogWell-Bench v1 Release	0	0	6	6										
	Total:	**36**	**36**	**36**	**108**										

3 Risks and Mitigation Strategies

Each risk is rated by likelihood (L) and impact (I) on a three-point scale (Low/Med/High), with its mitigation.

ID	Risk	L	I	WP	Mitigation
R1	Finite interaction data makes probe-based $\mathcal{V}$-information diagnostics unreliable.	Med	High	WP1	Calibrate on pilot data; fall back to proxy diagnostics (unassisted accuracy, help-seeking, transfer).
R2	Regime misclassification triggers inappropriate interventions.	Med	High	WP1	Calibrated uncertainty; require repeated evidence before major policy shifts.
R3	Interaction logs may not reveal latent cognitive traits or states.	Low	Med	WP2	Align profiles with psychometric instruments; interpretable concept-bottleneck models; report identifiability limits.
R4	Personal memory grows too large or retrieves irrelevant context.	Med	Med	WP2	Structured insight consolidation; abstractive forgetting; utility-ranked retrieval.
R5	Inaccurate UWM simulations mislead planning and self-play.	Med	High	WP3	Reliability self-diagnostics; switch to full model or observed signals when uncertain.
R6	$\mathcal{V}$-information gain may not track genuine long-term capability (proxy misalignment).	Med	High	WP4	Correct shaped reward with delayed grounded outcomes; recalibrate probe each cycle.
R7	Probe-diagnosed credit assignment is too noisy to outperform outcome-only training.	Med	Med	WP4	Redistribution flattens toward uniform interval-level credit; Reduces to NOVER-style verifier-free outcome filtering.
R8	Trial attrition or under-recruitment reduces statistical power.	Low	High	WP5	Phased cohort rollout; recruitment buffers; engagement monitoring; pre-registered analyses robust to missing data.
R9	Personalised AI increases over-reliance, distress, or inequity.	Med	High	All	Early ethics approval; dependency and distress monitoring; safety safeguards; equity analyses across learner groups.

Appendix: All ongoing and submitted grants and funding of the PI (Funding ID)

Mandatory information (does not count towards page limits)

Current research grants (Please indicate "No funding" when applicable):

Project Title	*Funding Source*	*Amount (Euro)*	*Period*	*Role of the PI*	*Relation to current ERC proposal*
AI-enabled High Stakes Assessment	EPSRC	705,437	2025–2028	PI	No scientific overlap. The project focuses on developing explainable AI marking tools for GCSE/A-level exams, not longitudinal human–AI co-evolution. It provides access to psychometric expertise that will inform WP1 probe-task design.
The EMBRACE study	Inkfish Research Foundation	40.62M	2025–2031	Co-I (AI theme lead)	Develops a personalised AI health assistant for maternal care. No scientific overlap.
AI Hub for Causality in Healthcare AI with Real Data	EPSRC	12M	2024–2029	Co-I	No scientific overlap. CHAI's causal-inference expertise informs WP5's longitudinal trial design and treatment-effect estimation.

On-going / submitted grant applications (Please indicate "None" when applicable):

Project Title	*Funding Source*	*Amount (Euro)*	*Period*	*Role of the PI*	*Relation to current ERC proposal*
Next-Generation Collaborative Agentic AI to Support European Strategic Industry Sectors	Horizon Europe	15M	2028–2031	Co-I	Develops multi-agent AI systems for industry applications. No scientific overlap.
Diffusion or Dilution? Tracing How AI Reshapes Scientific Knowledge Across Disciplines	EPSRC	232,100	2027–2028	PI	Investigates how AI-enabled scientific discovery reshapes the structure and quality of scientific knowledge. No scientific overlap

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